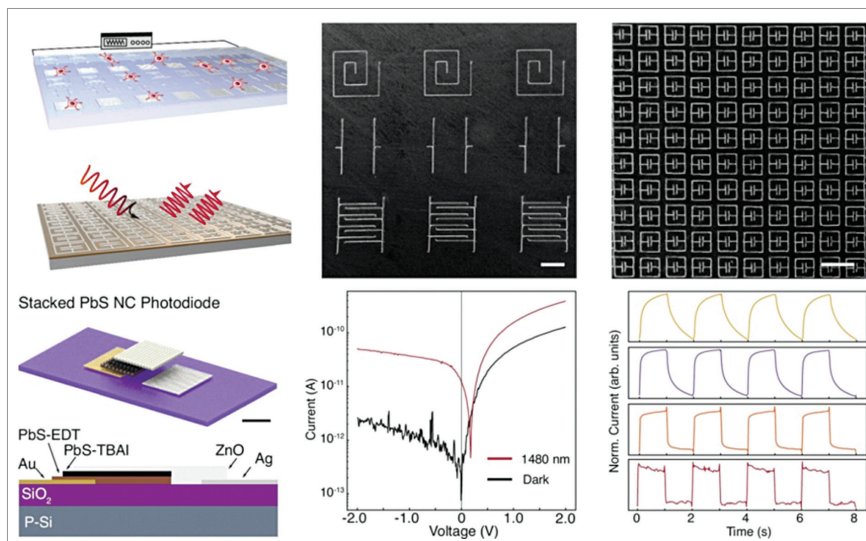


Printed infrared sensors break size limits



Printed microelectrode and nanoresonator arrays and all-printed infrared photodiodes
(Credit: The University of Hong Kong)

The world's smallest fully printed infrared (IR) photodetectors were developed by engineers at the University of Hong Kong, advancing miniaturisation beyond the limits of silicon. Using a novel room-temperature nano-printing platform that integrates colloidal nanocrystal inks with precision patterning, the team produced sub-10 μ m IR sensors, promising compact, low-cost integration in electronics. The technology could boost near-IR applications in autonomous systems, biomedical diagnostics, and optical communications. Unlike traditional silicon approaches requiring costly material bonding, this all-printed method supports scalable, multi-material designs.

Hidden climate cost of health wearables

The rapid growth of wearable health devices could carry a substantial hidden climate cost, a new study warns. Researchers from the University of Chicago and Cornell University estimate that devices such as glucose monitors, ECG patches, and blood pressure trackers emit 1.1-6.1kg of CO₂-eq each over their lifecycles. With global adoption rising, annual emissions could reach 3.4 million metric tons by 2050, driven mainly by PCB and semiconductor manufacturing.

Conceptual illustration of a wearable health care electronics network, where interconnected body-worn electronics enable continuous physiological monitoring, human-computer interaction, and real-time health care supports (Credit: Nature 2025)

